

GitHub Copilot Advanced Features – Demo Guide

Explore the capabilities of GitHub Copilot that go beyond code completion. Use this guide to showcase the most impactful features in your next demo session.

✨ 1. AI-Powered Code Suggestions

Best for: Demonstrating seamless in-editor productivity boosts.

- **Inline Code Completions:** Real-time suggestions for code, comments, tests, and more.
- **Partial Acceptance:** Accept only the next word or next line instead of the full suggestion.
- **Alternative Suggestions:** Choose from multiple completions for flexibility.
- **Suggestions from Comments:** Write a comment like `// generate a for-loop` to trigger Copilot guidance.
- **Refactoring Help:** Rename variables project-wide or adjust copied code to match style.

2. Natural Language Chat

Best for: Showing intuitive interactions between developer and Copilot.

- **Multi-turn Chat:** Ask questions, get follow-ups, stay in context.
- **Ask Mode:** Learn about your codebase, concepts, or general coding topics.
- **Edit Mode (Copilot Edits):** Describe code changes in plain English—Copilot edits across files.
- **Voice Interactions:** Dictate prompts via voice for hands-free coding.
- **Chat History Management:** Track and reuse previous sessions.

3. Agents & Multi-Step Code Planning (★)

Best for: Showing Copilot's ability to *think, plan, and act autonomously*.

Agent Mode

- Define high-level goals like *“create a new python flask app that can generate QR codes in the py_website folder. make sure to structure your folders correctly”*.
- Copilot creates and executes a plan across files and tools.
- Automatically invokes edits, runs commands, and iterates.

Custom MCP Tools & Agents

- Build your own tools Copilot Agents can use.
- Extend functionality for unique workflows or tech stacks.

 *Tip: Show off an Agent completing a multi-file task from a single natural language prompt.*



4. Context Awareness & Prompt Control

Best for: Tailoring suggestions to real-world codebases.

- **Automatic Context Discovery:** Use `#codebase` to auto-include relevant files.
- **Manual Context Inclusion:** Use `#file`, `#selection`, `#problems`, etc.
- **Fetch Web Content:** Use `#fetch` to pull in context from a URL.
- **Workspace Indexing:** Fast search across remote or local projects.

5. Smart Actions in the Editor

Best for: Productivity-focused demos inside VS Code.

- **Generate Commit Messages**
- **Pull Request Titles & Descriptions**
- **Rename Symbols with Suggestions**
- **Generate Inline Docs**
- **Generate Tests** (unit, integration, end-to-end)
- **Code Reviews** (quick or deep)
- **Semantic Search**: Find meaning-based matches, not just text.
- **Scaffold Projects/Files**: Use `/new` to spin up new code quickly.
- **Generate Notebooks**: Use `/newNotebook` to create Jupyter notebooks.

6. Slash Commands & Chat Enhancements

Best for: Fast, task-specific Copilot interactions.

- `/explain` , `/fix` , `/tests` , `/startDebugging` , `/new` , etc.
- Use **chat variables** like `#file` , `#selection` , or `#VSCodeAPI` to add structured context.
- Interact with **chat participants**:
 - `@workspace` – Ask about files/project
 - `@vscode` – Ask about VS Code settings/extensions
 - `@terminal` – Get help with shell commands
 - `@github` – PRs, issues, workflows



7. Experimental & Extensible Features

Best for: Showing what's possible with advanced or team-specific setups.

- **Custom Instructions:** Configure in `.github/copilot-instructions.md` to match team style.
- **Reusable Prompt Files:** Store task-specific prompts in `.prompt.md` to reuse/share.
- **Language Model Selection:** Choose from fast/lightweight or more advanced LMs.
- **Extend with APIs:**
 - Build custom actions, tools, and chat integrations.
 - Integrate Copilot deeply into your dev workflow.

8. Debugging & Fixes

Best for: Showing how Copilot can help after things break.

- **Debug Config Suggestions**
- **Inline Fixes** with `/fix` or Code Actions
- **Help with Test Failures or Lint Errors**



9. Vision (Preview)

Best for: Advanced demos or future-looking capabilities.

- Add screenshots/images to chat prompts.
- Ask questions about visual content (great for UI bugs).